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Higher national Diploma in Accountancy

Working Capital Management and Firm's Performance of Manufacturing Companies in Sri Lanka

Group 01

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Abbreviations

1. WCM – Working Capital Management
2. ROA – Return on Assets
3. ROE – Return on Equity
4. CCC – Cash Conversion Cycle
5. CSE – Colombo Stock Exchange
6. ICP – Inventory Conversion Period
7. DCP – Debtors Conversion Period
8. CDP – Creditors Deferral Period
9. GICS – Global Industry Classification Standard

CHAPTER ONE

Introduction

1.1 Background of study

Working capital management (WCM) is vital for maintaining the financial health and operational efficiency of manufacturing firms. Effective WCM ensures balanced management of current assets and liabilities, enabling firms to sustain operations, maintain liquidity, and maximize profitability. In Sri Lanka, where the manufacturing sector plays a significant role in economic growth and employment, understanding the impact of WCM on firm performance is crucial especially considering recent economic challenges.

WCM involves managing short term assets and liabilities, such as inventories, receivables, payables, and cash, to ensure business continuity and financial stability. Efficient WCM helps prevent operational disruptions, reduce financing costs, and optimize resource use. Finance theory and empirical research confirm that sound WCM significantly influences profitability and firm value (Delima, 2020; Murigu, 2019).

Striking a balance between liquidity and profitability is complex. Poor management of WCM components can threaten a firm's sustainability, especially in capital-intensive sectors like manufacturing. The COVID-19 pandemic further emphasized the importance of flexible and efficient WCM strategies. Sri Lankan firms faced cash flow issues, delayed payments, and supply chain disruptions, which challenged their operational strength (Weerasinghe & Perera, 2020; Silva & De Silva, 2021). Firms with adaptive WCM approaches, supported by real-time data and forecasting, managed these shocks more effectively (Perera, 2022).

While studies from developed economies have demonstrated that efficient WCM enhances profitability and shareholder value (Ukaegbu, 2015; Afrifa, 2016; Nazir & Afza, 2018), research in Sri Lanka remains limited. Recent studies have shown that inventory and receivable management, as well as shorter cash conversion cycles, positively influence profitability (Wijekoon & Azeez, 2021). However, the impact of individual WCM components on the financial performance of Sri Lankan manufacturing firms is still underexplored.

The Sri Lankan manufacturing sector, a key contributor to GDP, faces unique issues such as inflation, currency depreciation, and supply chain instability. Efficient WCM can help firms navigate these challenges by improving liquidity, enhancing profitability, and supporting business continuity. Empirical evidence suggests that shorter cash conversion cycles and better inventory and payment practices lead to stronger financial performance (Dissanayake, 2023; Wijesiri & Gunawardena, 2022).

Internationally, studies from Ghana and India also affirm that efficient WCM practices enhance return on assets (ROA) and return on equity (ROE), highlighting the relevance of regional economic contexts (Amponsah-Kwatiah & Asiamah, 2021). In Sri Lanka, findings indicate that both excessive and insufficient working capital negatively affect profitability, emphasizing the need for optimal WCM levels (Rajeev, 2017).

Challenges persist, particularly for smaller firms and those with high leverage, which face difficulties accessing finance and maintaining operational flexibility (Ahmed & Javed, 2020). Thus, a balanced and strategic approach to WCM is essential.

As Sri Lanka continues to face economic volatility, this study aims to explore the impact of WCM components on the financial performance of listed manufacturing companies. The findings will offer valuable insights for businesses, policymakers, and academics, and help develop best practices suited for dynamic and uncertain economic environments.

1.2 Statement of problem

The manufacturing sector plays a vital role in Sri Lanka’s economy, contributing significantly to GDP, employment, and exports. However, after COVID 19, with ongoing economic fluctuations, rising inflation, and supply chain disruptions, firms face increasing challenges in managing their working capital efficiently. Proper working capital management ensures liquidity, reduces financial risks, and enhances profitability, making it a critical factor in sustaining business operations. Effective working capital management is crucial for manufacturing firms' financial health and operational efficiency. An optimal balance in current asset investment is essential to avoid the risks associated with both excessive and inadequate working capital, as defined by Ehiremmen (2017).

Overinvestment in current assets may lead to inefficiencies, as idle funds generate no returns, negatively impacting profitability. According to Aminu and Zainudin (2015), Working Capital Management is considered the management of the firm's current assets and current liabilities. Conversely, underinvestment in working capital can threaten a firm's solvency by impairing its ability to meet short term obligations. The dynamic nature of business activities further complicates working capital requirements, necessitating continuous adjustments to maintain financial stability. In manufacturing firms, working capital management also involves securing sufficient funds to finance short term assets. Financial managers must constantly monitor and manage current asset levels, as fluctuations can significantly impact profitability. Failure to maintain optimal working capital levels can result in liquidity constraints, increased borrowing costs, and diminished firm value. Empirical studies in developed economies have established that efficient Working capital management improves firms' performance. Abdullahi (2020) finds that working capital management has a positive and significant effect on firm performance, indicating that efficient working capital management can improve a firm's financial performance.

However, few studies in developing countries are in this direction. In Sri Lanka, few attempts have been made to verify at the empirical level, the working capital management and firms' performance of the manufacturing sector in Sri Lanka. Mithushana, Kevin & Anandasayanan. (2019) focused on the relationship between working capital management and firm performance in listed manufacturing companies in Sri Lanka. To address this gap, I intend to conduct a study on how working capital management affects firms' performance of manufacturing companies in Sri Lanka.

1.2.1 Research problem

How Working Capital Management affects firms' performance of Manufacturing Companies in Sri Lanka.

1.3 Research Questions

Effective working capital management is crucial for enhancing profitability and ensuring the sustainable growth of manufacturing firms in Sri Lanka. This study aims to investigate the various aspects of working capital management and its impact on financial performance, operational efficiency, and strategic adaptation in the manufacturing sector. By addressing these aspects, the research seeks to provide valuable insights into the role of working capital management in sustaining business success. To achieve this objective, the study is guided by the following key research questions:

1. How does working capital management impact the financial performance of manufacturing firms in Sri Lanka?

2. What are the key components of working capital that influence firm profitability?

3. How do manufacturing firms in Sri Lanka currently manage their working capital?

4. What challenges do these firms face in maintaining optimal working capital levels?

1.4 Research Objectives

This study aims to analyze the effect of working capital management components on the performance of Sri Lankan listed manufacturing companies on the Colombo Stock Exchange (CSE). To achieve this primary objective, the study focuses on the following sub-objectives:

1. To examine the impact of working capital management on the financial performance of manufacturing firms in Sri Lanka.

2. To identify the key components of working capital (inventory, receivables, payables, etc.) that significantly influence firm performance.

3. To analyze the working capital management practices used by manufacturing firms in Sri Lanka.

4. To explore the challenges faced by manufacturing firms in maintaining efficient working capital levels.

1.5 Hypotheses Development

H1: There is a significant impact of working capital management on the financial performance of manufacturing firms in Sri Lanka.

H2: There is a significant relationship between inventory turnover period and firm profitability and liquidity.

H3: There is a significant relationship between accounts receivable collection period and firm profitability and liquidity.

H4: There is a significant relationship between account payable deferral period and firm profitability and liquidity.

1.6 Significance of study

This study examines the impact of working capital management (WCM) on the financial performance of listed manufacturing companies in Sri Lanka, using data from the past five years a period marked by economic instability. The economic crisis since 2019, characterized by high inflation, currency depreciation, and shortages of essential goods and foreign exchange, has significantly affected manufacturing operations.

The manufacturing sector is a key contributor to Sri Lanka's economy, accounting for approximately 17.92% of GDP in 2023. In this context, efficient WCM is critical for maintaining liquidity, meeting short-term obligations, and capitalizing on growth opportunities.

This research provides timely insights into effective WCM strategies during economic downturns. The findings will offer practical guidance for managers aiming to improve operational efficiency and financial performance, while also informing policymakers and investors working to support the sector's recovery and growth.

By incorporating current data and a broader focus on the manufacturing sector, this study offers updated and practical insights. The results will help firms optimize key WCM component inventory, receivables, and payables to enhance profitability and strength. It also aims to support the development of effective financial strategies to strengthen Sri Lanka's manufacturing sector and its contribution to economic growth.

1.7 Scope of study

This study focuses on assessing the impact of working capital management (WCM) on the financial performance of listed manufacturing companies in Sri Lanka. The manufacturing sector is a vital part of the national economy, contributing approximately 17.92% to GDP in 2023. It is particularly relevant due to its significant reliance on working capital investments, especially in inventory, receivables, and payables. Efficient management of these components is essential for maintaining liquidity, profitability, and operational stability, particularly during periods of economic volatility.

The study covers a five year period from 2020 to 2024, encompassing the recent economic crisis that began in 2019. This period was marked by high inflation, currency depreciation, shortages of essential goods, and foreign exchange constraints, all of which significantly disrupted manufacturing operations. Analyzing this timeframe allows the study to capture firm responses and adaptations to both pre-crisis and crisis conditions.

The research is limited to listed manufacturing firms due to the availability of consistent and reliable financial data through annual reports and stock exchange disclosures. While SMEs and firms in other sectors also contribute to the economy, they are excluded due to data limitations. As such, the findings reflect the experience of publicly listed manufacturers.

Key WCM components analyzed include inventory management, accounts receivable, and accounts payable. The study explores their individual and collective impact on profitability and operational efficiency, with particular emphasis on the cash conversion cycle as a measure of WCM efficiency.

This study uses secondary data obtained from financial statements, annual reports, and Colombo Stock Exchange (CSE) disclosures. Primary data collection methods such as surveys or interviews are not included, as the research emphasizes quantitative financial analysis based on verifiable historical data.

In summary, this study examines how WCM affects the financial performance of listed manufacturing firms in Sri Lanka from 2020 to 2024, using secondary data. It offers valuable insights for managers, policymakers, investors, and analysts aiming to

enhance financial resilience and operational efficiency in the manufacturing sector amid ongoing economic challenges.

1.8 Chapter Framework

This study comprises five chapters, each addressing a key aspect of the research. The first chapter introduces the study, including the background, problem statement, research questions, objectives, and significance of the research. Chapter two presents a comprehensive literature review, discussing both theoretical and empirical evidence from previous studies relevant to the topic.

The third chapter focuses on research methodology, detailing the data collection process, sample selection, hypotheses, conceptualization, operationalization, and analytical methods used in the study. Chapter four is dedicated to data collection and analysis, where information is gathered from annual reports, analyzed, and presented through various figures and graphs.

Additionally, this chapter includes hypothesis testing. Finally, chapter five summarizes the study's key findings and conclusions while also offering recommendations for future research.

1.9 Limitations of Study

The researcher identifies the following limitations in this study.

1. The analysis of this study is mainly based on secondary data.

2. The study covers 5 years from 2018/2019 to 2023/2024.

3. The financial statements are generally based on historical or original cost. The current economic condition is ignored.

4. Only the printed data about the company will be available and not the back-end details.

CHAPTER TWO

Literature Review

2.1 Introduction

A literature review is an essential part of any research project, fulfilling several important roles that improve the study's overall quality and accuracy. It offers a detailed summary of existing knowledge on a topic, enabling researchers to understand the current state of the field and pinpoint key themes, trends, and gaps in the literature. This integration of previous research not only positions the new study within the wider academic landscape but also underscores the importance of the research question being examined.

Given the increasing challenges faced by firms in Sri Lanka, particularly in the wake of economic crises and the COVID 19 pandemic, understanding the relationship between Working Capital Management (WCM) and firm performance has become more critical than ever. This study aims to bridge the existing gap in literature by focusing on how businesses in Sri Lanka's manufacturing sector have adapted their WCM strategies to navigate financial instability and ensure sustainability. By examining these strategies, the research seeks to offer valuable insights into the role of WCM in maintaining profitability, liquidity, and overall firm performance during periods of economic instability.

2.2 Concept review

2.2.1 Working capital management

Working capital is defined as the difference between a company's current assets and current liabilities, serving as a measure of liquidity for daily operations. It plays a crucial role in financial management, reflecting a firm's capability to meet short-term obligations and support its operational needs. Sufficient working capital is vital for ensuring seamless business operations, allowing a company to manage its expenses, invest in inventory, and address unforeseen financial issues. (Arunkumar, O.N. and R.T. Radha, 2017).

Working capital management (WCM) is essential for ensuring a firm's liquidity, operational efficiency, and overall profitability. The relationship between WCM and

firm profitability has received considerable focus in financial research. This literature review consolidates existing studies on the connections between WCM and firm financial performance. Maximization of the business profitability is an effect of working capital management, but also the reverse causality is plausible when firms are profitable, they have more cash to invest in working capital (Anton & Afloarei, 2021).

2.2.2 Components of working capital management

Grasping working capital's many dimensions is vital to understanding the idea. Gross and net working capital are the two main categories. Gross Working Capital (GWC) is: This phrase describes all a company's current assets. All liquid assets utilized for daily operations and short-term debt settlement are included in GWC. A company's current assets might occasionally account for a large amount, even half of its overall assets. Gross working capital refers to a company's investment in current assets including cash, bank deposits, short term securities, accounts receivable, and inventories.

The difference between current assets and short-term liabilities is known as Net Working Capital. It shows the amount of cash on hand to cover everyday expenses and takes advantage of profitable investment opportunities. Accordingly, working capital is frequently seen as a company's lifeblood, and while effective management can ensure success, ineffective management can result in bankruptcy (Yadav, Kamath, and Manjrekar, 2019).

WCM aims to establish an optimal level of working capital, continually monitoring and controlling individual components to prevent exceeding this ideal level, while also providing the necessary funds to support current assets. As an independent variable working capital management plays a major role in the organization. The following are the key factors we need to analyze to support our research. Those are the Inventory Conversion Period, Debtors Conversion Period, Creditors Deferral Period & Cash Conversion Cycle.

2.2.2.1 Inventory conversion period

The inventory turnover period is a financial measure that assesses how effectively a company manages its inventory by calculating the average time it takes to sell that inventory within a given timeframe. This metric is essential for evaluating a firm's operational efficiency and liquidity; a shorter inventory turnover period suggests that a

company is quickly selling its inventory, reducing the likelihood of encountering issues related to excess stock or obsolescence. In contrast, a longer inventory turnover period may indicate inefficiencies, overstocking, or decreasing sales.

Research indicates that achieving an optimal inventory turnover period is crucial for improving profitability and cash flow. For example, Gupta and Singh (2021) found that companies with higher inventory turnover ratios typically enjoy better financial performance, as they can reinvest cash into operations more swiftly. Similarly, a study by Wang, (2020) explored the link between inventory turnover and profitability in the manufacturing sector, concluding that effective inventory management plays a significant role in a firm's overall profitability. Existing research highlights its significance in driving profitability and cash flow, underscoring the necessity for businesses to monitor and refine their inventory turnover strategies closely.

2.2.2.2 Debtors conversion period

The Debtors Conversion Period (DCP) is a financial indicator that quantifies the average number of days it takes for a company to collect payments on its accounts receivable. This metric is essential for evaluating how effectively a business manages its credit and debt collection processes. It offers valuable insights into cash flow, liquidity, and the overall financial condition of the company. To determine the DCP, the formula is usually presented as:

DCP = (Account Receivable / Credit Sale) x 365 Days

Research indicates that a suitable DCP can differ significantly across industries. For instance, in the retail sector, where transactions are generally completed swiftly, the DCP is often much shorter compared to industries such as manufacturing, where payment delays are more common. A lengthy DCP may signal potential cash flow challenges, prompting the need for adjustments in credit policies or collection tactics (Anton, 2021). Additionally, analyzing customer payment patterns and categorizing them can enhance collection efforts, leading to better DCP. The Debtors Conversion Period is a crucial metric that affects a company's cash flow and overall financial stability.

2.2.2.3 Creditors deferral period

The Creditors Deferral Period (CDP) is a financial indicator that reflects the average number of days a company takes to settle its accounts payable and pay its suppliers. This metric is crucial for evaluating a business's cash flow management, liquidity, and its interactions with creditors. An effectively managed CDP can signify wise credit usage while fostering strong supplier relationships, which are essential for operational efficiency. To compute the CDP, the formula is generally represented as:

CDP = Average Accounts Payable * Days in Period / Total Credit Purchases.

Research indicates that the ideal CDP can differ greatly by industry. For instance, retail companies may have shorter CDPs because of the need for quick inventory turnover, while manufacturers might experience longer periods due to the intricacies of their supply chains. A longer CDP could signal cash flow issues or suggest that a company is skillfully managing its cash by delaying payments without harming supplier relationships Wang, (2020). Proper management of the CDP can lead to balanced cash flow, enhancing the company’s capacity to invest in growth and navigating financial challenges.

2.3 Working capital management firms' performance

Mithushana, Kevin & Anandasayanan, (2019) focused on the relationship between working capital management (WCM) and firm performance in listed manufacturing companies in Sri Lanka. Their research underscored the importance of maintaining an optimal cash conversion cycle (CCC) to enhance profitability. They emphasized that excessive investment in current assets could lead to lower returns, while inadequate investment might increase liquidity risks. Therefore, striking a balance in working capital components is essential for sustaining financial performance.

In a similar, Akingunola, Ajibola & Adigun (2020) investigated the relationship between WCM and firm performance and found that WCM has a significant effect on firm performance. Abdullahi (2020) further supports this by concluding that efficient WCM positively impacts a firm's financial performance, suggesting that effective management can lead to improvements in profitability.

Balogun and Adeyeye (2021) also found that WCM has a positive and significant effect on firm performance, suggesting that managing working capital efficiently is crucial

for improving a firm's financial health. Aminu and Zainudin (2015) describe WCM as the management of a firm's current assets and current liabilities, with four key components: Inventory Management, Accounts Receivable, Accounts Payable, and Cash Conversion Cycle.

In contrast, Afrifa & Tingbani (2018) examined the relationship between CCC and firm performance (measured by Tobin's Q and ROA) within UK SMEs. They found that CCC is significantly and negatively related to firm performance. However, when the availability of cash flow was accounted for as a moderating variable, the relationship became positive and significant.

On the other hand, Jakpar & Johari (2017) studied the relationship between CCC and ROA in Malaysian firms and reported that CCC was not significantly related to ROA, indicating that WCM did not contribute to firm performance or profitability levels.

The ongoing shifts in the business environment, such as the inclination toward expansion, rising competition, advances in information and communication technology, and increased focus on quality and capacity building, make the enactment of an efficiency measurement system for WCM, particularly in the manufacturing sector, a prevailing topic in research (Chiang, Lee & Liao 2019).

2.3.1 Working Capital Management and Profitability

Rathnayake (2021) examined the impact of working capital management on profitability among listed Sri Lankan companies. Analyzing data from 41 firms over four years, the study found that the cash conversion cycle and accounts payable days had a significant negative impact on return on assets (ROA). Conversely, days of accounts receivable exhibited a significant positive effect on ROA. These findings suggest that efficient management of receivables and payables is critical for enhancing profitability. The study also highlighted that firm size positively influenced both profitability and firm value, indicating that larger firms might benefit from economies of scale in working capital management.

Nguyen, Manh-Dung & Duc-Trung (2016) examined the effect of working capital management on firms' profitability by using data from listed companies on the Vietnamese Stock Exchange. The result implies that there is no correlation between working capital management and firms' profitability.

Gambo & Abdulkarim (2016) recommended that management of food and beverage companies in Nigeria should reduce the period involved in their cash conversion cycle. This, they suggested, would improve the company's performance by increasing profit generation and making more free cash flow available to be distributed to shareholders at the end of the company's accounting period.

Abraham (2020) recommended that the average payment period has a significant positive relationship with profitability, while sales growth and firm size influence profitability in opposite directions. The study concludes that firms need to improve their collection and payment policy. Furthermore, firms can enhance profitability by identifying and concentrating on target markets, rather than simply focusing on increasing sales. Efficient management and financing of working capital components can increase the operating profitability of sole proprietorship manufacturing firms.

Golas (2020) conducted a study on the food industry sub sector, and the results showed that there is a statistically significant causative relationship between a firm's working capital and financial performance. Based on the developed regression model, it is evident that the increase in inventory days negatively impacts the return on operating assets. The analysis also proves the importance of accounting for the inventory mix, particularly the ratio of raw materials to work in progress, which has the greatest negative impact on profitability.

However, Kwatiah and Asiamah (2021) conducted research on working capital management and profitability of listed manufacturing firms in Ghana, using a sample of 20 listed firms for the period from 2015 to 2019. They found a strong positive and significant relationship between inventory management and return on assets (ROA) as well as return on equity (ROE). A similar result was obtained in their research, confirming the importance of working capital management for profitability.

2.3.2 Working Capital Management and Liquidity

Liquidity is a key element of working capital management, as it ensures a firm can meet its short term obligations. Liquidity is measured using the current ratio and the quick ratio. The current ratio is determined by dividing current assets by current liabilities, while the quick ratio is obtained by dividing current assets (net of inventory) by current liabilities. Under normal circumstances, a current ratio of 2:1 and a quick ratio of 1:1 is considered standards to indicate a sound liquidity position for a firm.

A study by Khan (2016) observed that Small Manufacturing Enterprises (SMEs) often operate below these standards, which increases their risk of facing liquidity and solvency challenges. An organization must maintain a balance between liquidity and profitability in its day to day operations, as failure to do so can lead to insolvency risks. He noted that “risk can be minimized by maintaining a higher level of current assets or working capital” and emphasized that firms that adequately plan their cash, inventory, and sundry debtors encounter fewer problems of control than those without effective policies in these areas.

Effective working capital management (WCM) can help ensure the smooth functioning and financial health of firms. Well managed working capital allows firms to generate higher profits without creating liquidity problems. While liquidity is a prerequisite for meeting short term commitments, firms with insufficient liquidity might face bankruptcy (Seth & Sharma, 2020). This direct effect arises because firms may need to borrow to finance working capital and cash gaps. For example, a firm with a cash gap of 50 days would need to borrow an amount equal to 50 times the cost of sales for daily operations. Such borrowing costs, rather than being used for operations, reduce pre- and post tax profits in equal proportions.

2.4 Empirical study

A substantial portion of the literature on working capital management (WCM) examines the relationship between WCM and a firm's profitability. These studies analyze the impact of WCM on overall firm performance, often using regression techniques. Various dependent variables related to profitability, such as return on assets (ROA), gross operating profit, and net operating profit, are typically considered. The primary independent variable used to gauge the efficiency of working capital management is the Cash Conversion Cycle (CCC).

Abdul Rahman and Parameshwara's (2022) study on Indian automobile manufacturers highlighted the importance of WCM in enhancing profitability. Their analysis of components such as inventory turnover ratio (ITR), receivables turnover ratio (RTR), and creditors turnover ratio (CTR) demonstrated that efficient management of these elements positively impacts the current asset turnover ratio, a proxy for profitability. For instance, Maruti Suzuki India Ltd. showed improved profitability with higher

inventory turnover, while Hyundai Motor India Ltd. benefited from reduced cash and other current assets.

Dissanayake & Mendis (2019) studied the food and beverage sector in Sri Lanka and found that efficient WCM positively influences firm performance, particularly ROA. Key components like Gross Working Capital (GTA), Current Assets to Total Assets Ratio (CATA), and Current Liabilities to Total Assets Ratio (CLTA) significantly affected profitability. A longer accounts collection period and slower inventory turnover were negatively correlated with ROA, while a higher ratio of current assets to total assets had a positive impact on profitability.

Kengatharan & Tissera (2019) investigated the relationship between corporate governance and working capital efficiency in Sri Lankan manufacturing companies. This study found that governance practices like CEO tenure and board meetings significantly influenced the cash conversion cycle. A shorter CCC, resulting from better WCM, was found to improve liquidity and overall firm performance.

Wang, (2020) explored the impact of WCM across different stages of the corporate life cycle in Pakistan. Their findings revealed that aggressive WCM strategies improved profitability during the introduction and growth stages but had diminishing returns during the maturity stage. This implies that firms at different stages of their lifecycle may need to adjust their WCM strategies for optimal performance.

Kasahun's (2020) study of sole proprietorship manufacturing firms in Ethiopia indicated that the average payment period positively impacts profitability, while the average collection period negatively affects it. The study emphasized the need for firms to improve their WCM policies to boost profitability.

Mandipa & Sibindi (2022) examined the retail sector in South Africa and found that the ACP and inventory turnover negatively affected profitability. Conversely, a longer accounts payable period had a positive but insignificant relationship with profitability. The study concluded that prudent WCM, particularly in managing accounts receivable and inventory turnover, could improve profitability.

Kafeel, (2020) used dynamic models to analyze the relationship between WCM and profitability in manufacturing firms. They found that the inventory conversion period (ICP) and creditors deferral period (CDP) positively affected ROA, while the CCC had

a negative impact on ROA. The receivable collection period (RCP) had a positive but statistically insignificant effect, suggesting that efficient inventory and payable management can enhance profitability.

Soukhakian & Khodakarami (2019) conducted a study on Iranian manufacturing firms, revealing a negative relationship between the CCC and both ROA and refined economic value added (REVA). They also found that inflation significantly impacts ROA, but macroeconomic factors like GDP did not moderate the relationship between WCM and firm performance. This emphasizes the importance of reducing the CCC to improve profitability, especially in volatile economic environments.

Zaman's (2022) study on manufacturing companies in Bangladesh concluded that company growth, profitability, and leverage significantly affect WCM. The research found that firms with higher profitability and better leverage tend to manage their WCM more efficiently, which in turn improves firm performance. This supports the notion that effective WCM can enhance a firm's financial health and profitability.

Mwangi, Kihara, & Muturi (2023) focused on the impact of WCM practices on the profitability of manufacturing firms listed on the Nairobi Securities Exchange (NSE). The study found that managing cash holdings, inventory, current liabilities, and debtors significantly influenced the profitability of firms. Efficient management of these components was linked to better financial performance, reinforcing the view that strategic management of short-term assets and liabilities is key to maximizing profitability.

Ahmad & Javed (2020) studied firms in Pakistan and showed that the inventory conversion period (ICP), debtors conversion period (DCP), and creditors deferral period (CDP) all negatively influenced ROA. The CCC had a mixed effect on firm performance, suggesting that reducing the CCC could improve profitability.

2.5 Research gap

Although numerous studies have explored the relationship between working capital management (WCM) and firm performance, the context of Sri Lanka presents a unique challenge, especially following the COVID 19 pandemic and the subsequent economic crisis. The existing literature predominantly focuses on pre pandemic data, failing to

account for the profound effects of these unprecedented events on manufacturing firms' financial health.

The economic fallout from the pandemic and the severe economic crisis that followed in Sri Lanka has significantly altered the financial landscape for many firms. Issues such as liquidity crunches, disrupted supply chains, and changes in consumer behavior have further complicated the efficient management of working capital. Research up to now has primarily assessed the impacts of WCM on profitability and liquidity in stable economic environments, with limited attention to the dynamic factors introduced by such crises.

There is a pressing need for empirical research that specifically investigates how working capital management influences firm performance during periods of economic instability. This study seeks to bridge this gap by analyzing how COVID 19 and the subsequent economic crisis have impacted the relationship between WCM and firm financial performance in Sri Lanka's manufacturing sector. Furthermore, it aims to understand how manufacturing companies have adjusted their working capital strategies to maintain profitability and liquidity under these challenging circumstances.

By exploring this gap, this research will contribute to the body of knowledge on WCM, particularly within the context of developing economies facing post-pandemic challenges. It will also provide practical insights for business managers and policymakers seeking strategies to navigate future economic uncertainties and improve firm performance in times of crisis.

2.6 Chapter Summary

This chapter provided a comprehensive overview of the existing literature concerning the relationship between Working Capital Management and Firms' performance. It involved a thorough examination of prior studies that are pertinent to this area of inquiry. The insights and findings gathered from this literature review served as the foundation for the current study. In the following chapter, we will delve into the research methodology, outlining the philosophical approach guiding the research, the sampling strategies employed, and the methods used for data collection.

Additionally, this chapter will detail how the variables were measured, ensuring a clear understanding of the analytical framework underpinning the study. This structured

approach will help clarify how the research was conducted and how the findings will be interpreted in relation to the existing body of knowledge.

CHAPTER THREE

Research Methodology

3.1 Introduction

Research methodology focuses on the research process and the kind of tools and procedures to be used. This chapter outlines research design, research approach, research method, study context, data collection, population and sampling, Conceptualization, Operationalization, Definition of variables, Hypothesis, Statistical tool, Model specification and Method of data analysis.

3.2 Research Design

This research will be descriptive studies. The emphasis here is on studying a situation or a problem to explain the relationship between variables.

3.3 Research Approach

An organized plan or technique for conducting research that directs data collection, analysis, and interpretation is called a research approach. It acts as a guide for the study's methodology and is essential to guarantee that the research is methodical, exacting, and reliable. As this study is business and management research based on secondary quantitative data, it adopts a positive research philosophy and primarily follows a deductive approach. This enables the testing of hypotheses about the relationship between working capital management and firm performance in listed manufacturing companies in Sri Lanka using existing financial data.

3.4 Sampling Techniques

At present, 284 companies are listed on the Colombo Stock Exchange and belong to 20 GICS industry groups. Stratified random sampling and random sampling techniques will be used to select the sample from the population. The stratified random sampling is used to select the sample from the proposed nature of the business. Therefore, eight manufacturing sectors are selected based on the nature of the business out of twenty sectors in the CSE. 38% of companies in the manufacturing sector will be randomly selected for the study purpose. Fifteen companies will represent in the sample out of one hundred thirty one companies in the sector.

3.5 Data Sources

The study will be used secondary data. This data will be collected from annual reports of selected companies. In addition, secondary data will be collected from research studies, books, journals, newspapers and ongoing academic working papers. The collected data will be processed and analyzed to make the study useful to the practitioners, researchers, planners, policy makers and academicians.

3.6 Variables

Working Capital Management is the process of managing a company's short-term assets and liabilities to ensure it can continue its operations without disruption. It involves making sure the company has enough cash flow to meet its day to day expenses and short-term obligations, like paying suppliers or employees. The Inventory Conversion Period (ICP) is the amount of time it takes for a company to sell its inventory and convert it into cash. It measures how efficiently a business is managing its inventory. A shorter inventory conversion period indicates that a company is selling and replenishing its inventory quickly, which is generally favorable for liquidity and profitability.

The Debtors Conversion Period measures the average number of days it takes for a company to collect payments from its customers after a sale has been made. This period indicates how efficiently a company is managing its receivables. The Cash Conversion Cycle (CCC) measures how long it takes for a company to convert its investments in inventory and other resources into cash flows from sales. In other words, it is the time taken to turn outflows (for raw materials, labor, and other costs) into cash inflows from customers. A shorter cash conversion period is generally preferred as it indicates efficient management of working capital.

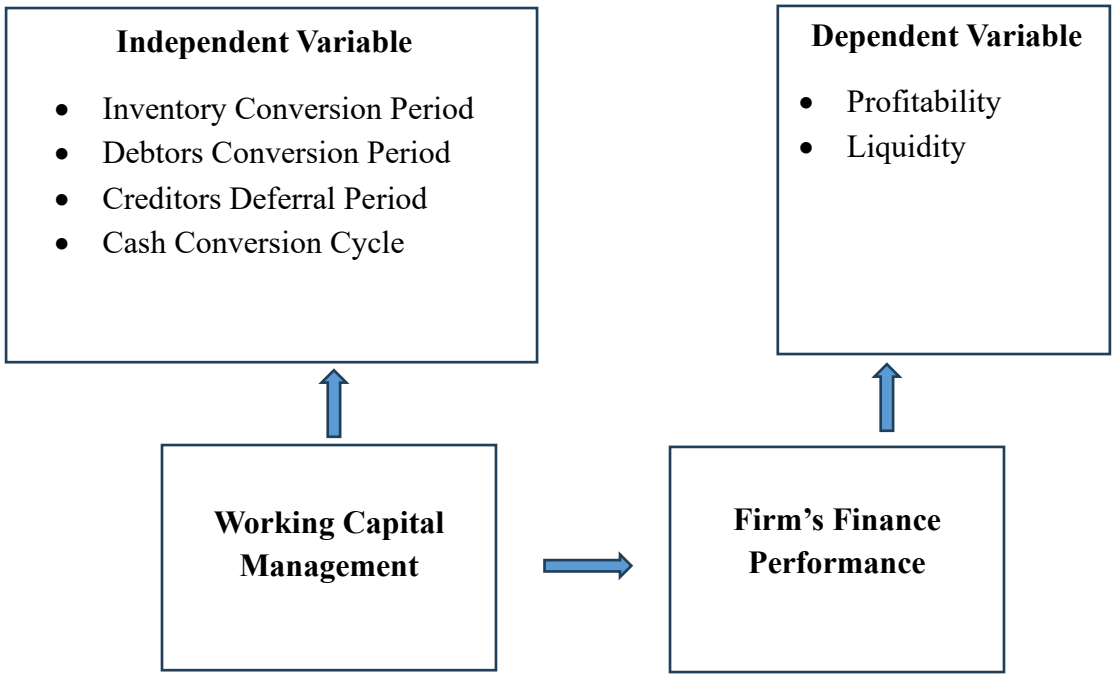
Liquidity refers to a company's ability to meet its short-term obligations or debts with its most liquid assets, such as cash or assets that can be quickly converted to cash. A company must have enough liquidity to cover its immediate operational costs (e.g., paying suppliers, employees, and other short-term liabilities). If liquidity is too low, the company might face cash flow problems and risk defaulting on payments. Profitability refers to a company’s ability to generate income relative to its revenue, assets, or equity. It's an indicator of how effectively the company is generating profit from its operations.

Profitability is crucial for the long-term survival and growth of the business. Profitable companies can reinvest in their operations, pay dividends, or accumulate reserves.

	Concepts	Variables
Independent Variable	Working Capital management	Inventory Conversion Period
		Debtors Conversion Period
		Cash Conversion Cycle
		Creditors' Deferral Period
Dependent variable	Firms Performance	Profitability
		Liquidity

3.7 Conceptual framework

Conceptual framework is the body of this research. It presents the relationship between independent variables and dependent variables. The research study is based on the following model, and it emphasizes the relationship between the working capital management and the firm performance of the organization. According to this conceptual framework represents the relationship between working capital management and firm performance. This study conducts to measure this conceptual framework relationship.



3.8 Operationalization

Concept	Variables	Indicators	Measures
Working Capital Management	ICP	Inventory, Cost of sales	(Inventory/Cost of Sales) X 365
	DCP	Trade receivable, Credit sales	(Receivable/Net credit sales) X 365
	CDP	Trade payable, Cost of sales	(Payable /Cost of sales) x 365
	CCC	ICP, DCP, CDP	CCC= ICP +DCP-CDP
Firm's Performance (Profitability)	ROA	Profit after tax, total asset	Net Income/ Total Asset
	ROE	Profit after tax, shareholders' equity	Net Income/ Shareholders' Equity
Liquidity	Current Ratio	Current Assets, Current Liabilities	Current Assets / Current Liabilities
	Quick Ratio	Current Assets, Inventory, Current Liabilities	(Current Assets - Inventory) / Current Liabilities
	Cash Ratio	Cash, Cash Equivalents, Current Liabilities	(Cash + Cash Equivalents) / Current Liabilities

3.9 Mode of Analysis

In the present study the data collected by descriptive statistics will be analyzed (i.e., means, maximum, minimum and standard deviation) inferential statistics (i.e pearson correlation and multiple regression). For the study, entire analysis will be done by personal computer. The results will be obtained by applying statistical tools, namely Stata.

Much simple action research involves descriptive analysis and provides valuable information about the nature of the group of samples (Best & Kahn, 2003). Therefore,

basic analysis for this study is carried out using descriptive statistical analysis. It shows central tendency and dispersion of the data consists of mean, minimum, maximum and standard deviation of each variable used in this research. Descriptive statistics allow researchers to continue other important analysis.

Correlation analysis is used to measure the strength of the association between the two variables. Value of coefficient (r) reveals features of association such as positive (+) or negative (-) and perfect (+1, -1) or weak ($0 < 0.3$ or $0 > -0.3$) or moderate ($0.3 < 0.7$ or $-0.3 > -0.7$) or strong ($0.7 < 1.0$ or -1.0) correlation. It shows individual relationship between only two variables when other variables are constant. In this study, it is used to analyze statistical relationship between selected variables in both theoretical models.

3.10 Chapter Summary

This chapter represented the objectives and methodology based on the research model and explained about the variables connected with the research. Furthermore, this chapter explains the research design, research sample, data collection techniques, data presentation and way of data analysis and interpretation followed by the researcher to conduct the research. The secondary data of the study were collected from the annual report of listed companies in CSE in the year of 2024.Researcher used their hypothesis to identify the relationship between independent variables and depended variables through correlations. Multiple linear regressions analyze the relative contribution of each factor.

CHAPTER FOUR

Data Presentation and Analysis

4.1 Introduction

This chapter includes data presentation and analysis. Presenting data involves the use of a variety of different techniques to visually show the reader the relationship between different data sets, to emphasis the nature of a particular aspect of the data. In this part researcher is going to discuss the results of the analysis. For presenting the data researcher going to use different statistical methods such as descriptive statistics, correlation analysis, regression analysis.

4.2 Descriptive Analysis for Independent and Dependent Variables.

Descriptive analysis is the important step in this analysis section as well as each research studies. It will help to describe and provide detailed information about each relevant variable. Mean, maximum, minimum, and standard deviation are the forms of descriptive statistics. In relevant to this research study descriptive statistics are calculating for ICP, DCP, CDP, CCC, ROA and ROE in selected Manufacturing companies in CSE by using suitable statistical package.

Table 4.1 Descriptive Analysis

	Mean	Median	Minimum	Maximum	Std. Dev	Obs
ICP	107.4054	75.6618	33.9668	242.8939	75.6193	75
DCP	83.2189	56.8035	9.7036	247.8066	58.1484	75
CDP	66.5199	66.4535	4.0826	164.7314	46.7501	75
CCC	124.1044	60.6852	1.1141	285.3469	81.9692	75
ROE	1.0941	0.2444	-0.1945	17.9737	2.6037	75
ROA	8.0956	0.0844	-0.0555	38.3087	7.4733	75
Current Ratio	2.39391	1.4056	0.0941	33.7094	4.0845	75
Quick Ratio	1.7574	0.9323	-0.3165	28.3368	3.4315	75
Cash Ratio	0.4537	0.1230	0.0104	9.7455	1.2082	75

Based on the statistics for the variables we can derive following insights, The average Inventory Conversion Period (ICP) is 107 days with a standard deviation of 75.62, ranging from 34 days to 243 days, indicating wide variation among firms. The Debtors

Conversion Period (DCP) has a mean of 83 days and standard deviation of 58.15, with minimum and maximum values of 10 and 248 days respectively. The Creditors Deferral Period (CDP) shows a mean of 67 days and a standard deviation of 46.75, varying between 4 and 165 days. The Cash Conversion Cycle (CCC) records a mean of 124 days, a standard deviation of 81.97, and values ranging from 1 to 285 days.

For profitability measures, the Return on Equity (ROE) shows a mean of 1.09, a standard deviation of 2.60, and ranges from –0.19 to 17.97. The Return on Assets (ROA) records a mean of 8.10 with a standard deviation of 7.47, and values between –0.06 and 38.31.

Regarding liquidity, the Current Ratio has a mean of 2.39, a standard deviation of 4.08, with values ranging from 0.09 to 33.71. The Quick Ratio shows a mean of 1.76, a standard deviation of 3.43, and ranges between –0.32 and 28.34. The Cash Ratio records a mean of 0.45, a standard deviation of 1.21, with a minimum of 0.01 and a maximum of 9.75.

4.3 Correlation Analysis

Correlation analysis is a statistical method used to assess a possible linear association between two continuous variables. Here results show the strength of association between variables. Correlation models are used to measure the degree of association between different variables under consideration. Correlation analysis is a tool to analyze the relationship between independent and dependent variables. It shows strong positive correlation, strong negative correlation, positive correlation and negative correlation for every variable.

Table 4.2 Pearson's Correlation analysis

Variable	ICP	DCP	CDP	CCC	ROE	ROA	Current Ratio	Quick Ratio	Cash Ratio
ICP	1.0000								
DCP	0.0514	1.0000							
CDP	0.4022	0.2008	1.0000						
CCC	0.4729	0.6686	-0.2320	1.0000					
ROE	0.2972	-0.0835	0.1118	-0.0576	1.0000				
ROA	0.3238	-0.0350	0.2482	0.1363	0.4676	1.0000			
Current Ratio	0.4383	0.3354	0.3096	0.4911	-0.0270	0.2843	1.000		
Quick Ratio	0.2386	0.3474	0.2973	0.4872	-0.0170	0.2709	0.9976	1.0000	
Cash Ratio	0.0516	0.1825	0.2969	0.3747	-0.0099	0.2392	0.9355	0.9385	1.0000

The analysis reveals that ICP has a moderate correlation with ROA (0.324) and weak correlation with ROE (0.297). This suggests that firms with longer inventory holding periods sometimes perform better, possibly because maintaining adequate stock helps them meet customer demand during supply disruptions or delays. However, it also indicates a potential trade-off between higher inventory carrying costs and improved customer satisfaction.

DCP shows negative and insignificant correlations with ROA (–0.035) and ROE (–0.083), indicating that extending credit to customers does not enhance performance and may increase the risk of bad debts. This aligns with the observation that Sri Lankan firms faced delays in collections during the crisis years, which strained cash flows. CDP demonstrates weak correlation associations with ROA (0.248) and ROE (0.112), suggesting that implying that delayed payments to creditors have a minor positive

influence on profitability. The cash conversion cycle (CCC) shows a weak positive correlation with ROA (0.136) and negligible correlation with ROE (0.058), reflecting a limited impact of cash conversion efficiency on returns.

The analysis shows that ICP has a moderate positive correlation with the current ratio (0.438) and weak correlations with the quick (0.239), indicating that higher inventory levels slightly improve liquidity. DCP and CDP show weak correlations with all liquidity ratios, suggesting that longer collection and payment periods modestly support liquidity. CCC has strong correlations with the current ratio (0.491), quick ratio (0.487), and moderate correlation with cash ratio (0.374), meaning firms with longer cash cycles tend to hold more liquid assets. Overall, WCM components are positively impact the profitability and liquidity of firms.

4.4 Regression Analysis

Table 4.3 Model 1 ROA

Variable	Coefficient	Std. Error	t-stat	p-value
ICP	0.0072942	0.0027291	2.67	0.016
DCP	0.0015804	0.0074972	0.21	0.833
CDP	0.0285037	0.0100599	2.83	0.005
Constant	0.1218201	0.0192484	6.33	0.000
F-statistic	15.47	R-squared		0.0675
Adjusted R-squared	0.0205	Prob		0.0015
Root MSE	0.1556	No. Obser		75

The regression analysis shows that ICP significantly improves ROA ($\beta = 0.0073$, $p = 0.016$), confirming that efficient inventory management enhances asset returns. CDP also has a positive and significant impact ($\beta = 0.0285$, $p = 0.005$), meaning that firms which extend supplier payments gain temporary profitability advantages. DCP remains insignificant ($p = 0.833$), suggesting that receivable collection efficiency does not significantly influence asset returns.

The model explains 6.7 percent of the variation in ROA ($R^2 = 0.067$). Although the explanatory power is weak, the F-statistic (15.47, p = 0.0015) confirms that the model is statistically significant overall. The weak explanatory power suggests that macroeconomic conditions such as inflation, exchange rate depreciation, and high interest rates may have overshadowed the role of WCM during the study period.

Table 4.4 Model 2 ROE

Variable	**Coefficient**	**Std. Error**	**t-stat**	**p-value**
ICP	0.0044587	0.0021976	2.03	0.044
DCP	0.000299	0.0011213	0.26	0.795
CDP	0.0107145	0.0047291	2.26	0.035
Constant	1.571905	0.4061269	3.97	0.000
F-statistic	11.58	R-squared		0.0937
Adjusted R-squared	0.0676	Prob		0.0012
Root MSE	0.7188	No. Obser		75

For ROE, both ICP (β = 0.0045, p = 0.044) and CDP (β = 0.0107, p = 0.035) are significant predictors, while DCP remains insignificant (p = 0.795). These findings confirm that inventory management and trade credit utilization are important in delivering shareholder returns.

The model explains 9.4 percent of the variation in ROE ($R^2 = 0.094$). Although still modest, this explanatory power is slightly higher than the ROA model. The model is statistically significant (F = 11.58, p = 0.0012). The consistent insignificance of DCP across both models implies that receivables management may not have been a managerial priority during the crisis years, with firms focusing more on inventory and payables to sustain operations.

4.5 Acceptance and Rejection of Hypothesis

Table 4.5 Hypothesis Testing Results

Hypothesis	**Accept / Reject**
H1: There is a significant impact of working capital management on the financial performance of manufacturing firms in Sri Lanka.	Accepted
H2: There is a significant relationship between inventory conversion period and firm profitability and liquidity.	Accepted
H3: There is a significant relationship between accounts receivable collection period and firm profitability and liquidity.	Rejected
H4: There is a significant relationship between account payable deferral period and firm profitability and liquidity.	Accepted

4.6 Chapter Summary

This chapter presented the analysis and findings of the study. Descriptive statistics were first carried out to summarize the key variables of working capital management and firm performance. Correlation analysis was then employed to examine the strength of relationships among variables, where results showed mixed associations with profitability and liquidity.

Regression analysis was conducted to identify the impact of working capital components on profitability. Finally, hypotheses were tested based on the results of major findings.

CHAPTER FIVE

Conclusion

5.1 Introduction

This chapter concludes this study by presenting the findings of the study, implications of findings, suggestions and recommendations; describes the limitations regarding this study which was faced by the research when the study is accomplished. Finally, directions for further research is given for future researchers.

5.2 Key findings of the Study

The key findings from studies on manufacturing companies in Sri Lanka highlight the critical role that working capital play in influencing the firm performance of these firms. Studies indicate that various aspects of working capital, such as ICP, DCP, CCP and CCC affect firm financial performance.

5.3 Summary of Major Findings of the Study

This study examined the impact of Working Capital Management (WCM) on the performance of listed manufacturing firms in Sri Lanka. The findings indicate that WCM does influence profitability and liquidity, but the effects are component-specific and shaped by the economic environment. The results show that the Inventory Conversion Period (ICP) had a significant positive relationship with both ROA and ROE, suggesting that efficient inventory practices supported profitability. This finding reflects the importance of maintaining adequate stock levels in the face of supply chain disruptions. In contrast, the Debtors Collection Period (DCP) was statistically insignificant, indicating that receivables management did not contribute meaningfully to firm performance. This may be due to weak enforcement of credit policies and the tendency of firms to extend credit to preserve sales during the crisis years.

The Creditors Deferral Period (CDP) was found to significantly improve profitability, highlighting the role of supplier credit as a vital source of short-term financing. However, reliance on delayed payments may have long-term risks for supplier relationships. Finally, the Cash Conversion Cycle (CCC) showed a positive association with liquidity but not with profitability, reinforcing the liquidity, profitability trade off. Overall, the study concludes that ICP and CDP are the key drivers of profitability in Sri Lankan manufacturing firms, while DCP plays a negligible role. The results also

underline that external macroeconomic conditions strongly influenced firm outcomes beyond WCM practices.

5.4 Research Questions Discussion

This section addresses the research questions formulated in Chapter One, integrating the findings from descriptive analysis, correlation analysis, and regression results.

5.4.1 Impact of WCM on Financial Performance

The evidence shows Working Capital Management (WCM) affects profitability, but the magnitude is modest. Both ROA and ROE models are statistically significant, yet with low explanatory power ($R^2 \approx 6.7\%$ - 9.4%), indicating external factors (e.g., inflation) also shaped outcomes. Within WCM, inventory conversion period (ICP) and creditors deferral period (CDP) significantly improved profitability, while debtors conversion period (DCP) did not, confirming a component-specific effect.

5.4.2 Key Components Influencing Profitability

Two components consistently explain profitability. The Inventory Conversion Period (ICP) is positively and significantly related to ROA and ROE, implying that maintaining efficient inventory supported operations and returns in the study period. The Creditors Deferral Period (CDP) is also positive and significant for both profitability measures, reflecting the role of supplier credit as short term financing. By contrast, the Debtors Conversion Period (DCP) is insignificant across models, indicating receivables management did not translate into higher returns.

5.4.3 Current WCM Practices of Sri Lankan Manufacturers

The descriptive results reveal wide dispersion across the key working capital measures Inventory Conversion Period (ICP), Debtors Collection Period (DCP), Creditors Deferral Period (CDP), and Cash Conversion Cycle (CCC) indicating that manufacturing firms in Sri Lanka do not follow a uniform approach to managing their short-term finances. Some firms operate with almost immediate cash conversion cycles, reflecting high operational efficiency, close integration of supply chains, and effective receivables collection. In contrast, others exhibit prolonged cycles extending over several months, which expose them to greater liquidity risk and increased financing costs.

The variation is equally pronounced in liquidity ratios. While certain firms maintain very high current and quick ratios, suggesting a cautious risk posture with substantial liquidity buffers, others operate with lean liquidity, reflecting either aggressive working capital policies or difficulties in building cash reserves during crisis years. The presence of extreme outliers, such as firms with unusually high current or quick ratios, also suggests potential over-investment in current assets. This can reduce profitability, as idle resources generate little return. Conversely, firms with very low cash ratios are heavily reliant on receivables and inventory to meet obligations, which may lead to financial stress in times of economic volatility.

Overall, these results highlight that Sri Lankan manufacturers adopt heterogeneous WCM practices, shaped by their size, bargaining power, and exposure to external shocks such as supply chain disruptions, inflation, and currency depreciation. This heterogeneity reflects differing managerial philosophies: some prioritize liquidity as a buffer against uncertainty, while others prioritize profitability through tighter working capital cycles.

5.4.4 Challenges in Maintaining Optimal WCM Levels

The study shows that Sri Lankan manufacturing firms face several challenges in managing working capital effectively. First, prolonged collection times are a major issue. Many firms take too long to collect payments from customers, which reduces available cash and creates pressure on day to day operations. Since customers also face financial difficulties, firms often struggle to enforce strict collection policies.

Second, many firms depend heavily on supplier credit. Delaying payments helps ease short term liquidity problems, but relying too much on this practice can damage supplier relationships and may lead to stricter terms or supply disruptions later on. Third, firms often hold large amounts of inventory. Keeping high stock levels protects against supply problems and market uncertainty, but it also increases storage costs and ties up funds that could be used more productively.

Finally, economic instability makes WCM more difficult. High inflation, currency depreciation, and import restrictions increase costs and disrupt cash flow planning. These external pressures force firms to focus more on survival than on optimizing profitability.

5.5 Conclusion

The purpose of this study was to examine the association between Working Capital Management and financial performance of listed manufacturing firms in Sri Lanka. Results validate that WCM does play some role in profitability and liquidity but is moderate in magnitude and varied in components. The analysis revealed that two elements of WCM Inventory Conversion Period (ICP) and Creditors Deferral Period (CDP) play the most important roles in shaping financial outcomes, while Debtors Conversion Period (DCP) and Cash Conversion Cycle (CCC) are less influential.

The significant and positive correlation between profitability and ICP indicates that sound inventory management is key to manufacturing firms. Firms with sufficient inventory to sustain them through uncertain periods managed to protect their returns. Similarly, CDP was strongly correlated with ROA and ROE, which indicate the use of supplier credit as a key source of short-term finance. This finding suggests that during periods of tightness in funding, corporations will rely significantly on delayed payments to ease liquidity pressure.

Contrary to this, there was no significant effect of DCP on performance. Consistent with Sri Lankan companies' struggles in getting receivables back from customers, particularly within a weak economic scenario where customers themselves are illiquid, DCP was related to liquidity but not profitability, confirming that longer cycles have the effect of making firms appear more liquid without necessarily enhancing returns.

By comparison, the findings reveal that Sri Lankan manufacturing firms' WCM is driven by needs rather than strategy. Inventory and supplier credit have become most important ways of maintaining operations and profitability, with receivable management remaining a weak link. Concurrently, the low explanatory capacities of the regression equations mean that external variables such as inflation, exchange rate volatility, and supply chain instability influence performance to a greater extent than WCM alone, but WCM is an important determinant of financial health and cannot guarantee profitability alone. To Sri Lankan businesses, the success of WCM relies on trading liquidity and returns, as well as practice adaptation to the economy's volatile nature.

5.6 Implication of the Study

The study adds to the literature by showing that the relationship of WCM with performance is not uniform across all its constituents. The research substantiates the contingency theory perspective, which believes that financial management results are contingent upon the context. Though ICP and CDP influenced profitability significantly, DCP did not have any influence, and CCC was linked more with liquidity than return. This would mean that WCM needs to be studied as a set of component-specific practices rather than a single construct. Also, the results highlight the liquidity, profitability trade-off, where increasing liquidity does not necessarily increase profitability. These findings extend theoretical understanding of WCM in emerging economies in crisis periods.

The results emphasize the importance of efficient inventory management and negotiation strategies with suppliers as both variables positively contribute to profitability. On the other hand, the lack of relevance of receivables management reflects an area of weakness that firms must work on. Managers must tighten credit collection systems through stricter policies, better monitoring, and use of technology such as electronic billing or automated reminders. In addition, firms should not over rely on supplier credit because it can negatively impact long term relationships. A balanced approach whereby liquidity is attained without tying up too much capital in current assets can maximize operating stability and shareholder wealth.

On a more general level, the findings call for policies that facilitate better WCM practices across the industry. The government and industry regulators may provide training and guidelines in receivables management, credit risk analysis, and inventory optimization. Access to low cost short term financing through banks or development funds would also deter companies from resorting to supplier credit. Encouraging digitalization of finance and accounting functions would also increase efficiency, particularly for receivables and cash management.

5.7 Limitation of the Study

First, the research relied solely on secondary data obtained from printed annual reports of quoted manufacturing firms. It limited analysis scope to quantitative financial indicators and excluded qualitative aspects such as managerial discretion, supplier

negotiation, or customer affairs, which may have been at work influencing working capital outcomes.

Secondly, the sample was confined to Sri Lankan listed manufacturing companies. While this ensured maximum reliability of data, it excluded small and medium enterprises (SMEs) and unlisted companies, where WCM practices may differ due to limited resources, positioning in the market, or governance structures. The outcomes, thus, cannot be extended to the entire manufacturing sector or any other sectors such as services or trading.

Thirdly, the time under review was five years (2020 - 2024), and this time coincided with unusual economic adversities, like the COVID 19 crisis, foreign exchange shortages, and political instability. These external events had a significant impact on firm performance and may have skewed the true impact of WCM under more favorable circumstances. Finally, the regression models had comparatively low explanatory power (R^2 figures less than 10%) and hence suggested that factors beyond WCM, such as macroeconomic policy, inflation, and firm-specific strategies, were more influential determinants of profitability.

5.8 Suggestion for future Research

Future research would benefit from increasing the sample size to include SMEs as well as unlisted companies. Smaller firms being more liquidity constrained, their working capital practices may exhibit other patterns than those observed with the listed companies. Research could employ the longer time horizon that covers both periods of stability as well as crisis periods. This would allow value chain management practice effects on performance across various economic conditions to be contrasted as well as the value of some strategies during expansion compared to downturn.

Next, adding qualitative techniques like interviews or questionnaires with financial managers would allow further insight into the rationale behind WCM choices, complementing the financial statement quantitative evidence. Finally, subsequent research could explore the effect of moderating and mediator variables such as firm size, ownership, governance, or access to finance. They could also identify the point up to which WCM converges into profitability. Industry based comparative research would establish whether the results obtained here are idiosyncratic to the economic environment there or a part of the larger pattern among the emerging economies.

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